

Crafting the Ultimate Docker Image for Spring Applications



bellsoft

whoami



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- Pasha Finkelshteyn



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- Dev 🥑 at BellSoft



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BellSoft

- Vendor of Liberica JDK
- Contributor to the OpenJDK
- Author of ARM32 support in JDK
- Own base images
- Own Linux: Alpaquita

Liberica is the JDK officially recommended
by 



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We know our stuff!



So, what is ultimate?

- Smallest
- Fatsest startup
- Something inbetween

So, let's look at all of
them!

How do we create an image?

Let's start from trivial.

```
1 FROM bellsoft/liberica-runtime-container:jdk-21.0.3_10-musl
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3 COPY . /app
4 RUN cd /app && ./gradlew build
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1 FROM bash
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4. We would like image to be light, but it's not :(

Our Dockerfile

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1	ID	TAG	SIZE	COMMAND
2	cda5201e70	dumb:latest	0B	CMD ["/bin/sh"
3	74a0788411		512.03MiB	/bin/sh -c cd /
4	7b82eb08ec		67.41MiB	COPY dir:ef251b
5	e4bcf83d29	bellsoft/liberica-runtime-container:jdk-21.0.3_10-musl	98.32MiB	15 BUNDLE_TYPE
6	<missing>		0B	ENV JAVA_HOME="
7	<missing>		0B	ENV LANG=en_US.
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579.44 MB are changed on every build!

Why do we care? We care because:

1. `push` takes longer time to start
(update is longer)
2. `pull` takes longer
(update takes longer & scaling takes longer)

Also, more disk pspace is inefficiently used



Optimizing. Round 1.

Let's build it outside of container

```
1 FROM bellsoft/liberica-runtime-container:jdk-21.0.3_10-musl
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4 RUN cd /app && ./gradlew build
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Just copying the prebuilt `jar` file

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5	e4bcf83d29	bellsoft/liberica-runtime-container:jdk-21.0.3_10-musl	98.32MiB	15 BUNDLE_TYPE=jdk CDS=n
6	<missing>		0B	ENV JAVA_HOME="/usr/lib/j
7	<missing>		0B	ENV LANG=en_US.UTF-8 LANG
8	<missing>		0B	LABEL org.opencontainers.
9	<missing>		0B	LABEL org.opencontainers.
10	<missing>		0B	LABEL maintainer="\$MAINTA
11	<missing>		0B	ARG BUNDLE_TYPE CDS DESCR
12	<missing>		0B	ARG BUNDLE_TYPE CDS JAVA_
13	<missing>		0B	ARG BUNDLE_TYPE CDS JAVA_
14	<missing>		0B	ARG BUNDLE_TYPE CDS JAVA_
15	<missing>		0B	ARG BUNDLE_TYPE JAVA_RELE
16	<missing>		0B	ARG JAVA_RELEASE
17	<missing>		7.49MiB	CMD ["/bin/sh"]

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1	ID	TAG	SIZE	COMMAND
2	cefb30e21d	smarter:latest	0B	CMD ["/bin/sh" "-c" "java
3	46d7594000		58.80MiB	COPY file:8d96bee95ae5ce43
4	e4bcf83d29	bellsoft/liberica-runtime-container:jdk-21.0.3_10-musl	98.32MiB	15 BUNDLE_TYPE=jdk CDS=no
5	<missing>		0B	ENV JAVA_HOME="/usr/lib/jv
6	<missing>		0B	ENV LANG=en_US.UTF-8 LANGU
7	<missing>		0B	LABEL org.opencontainers.i
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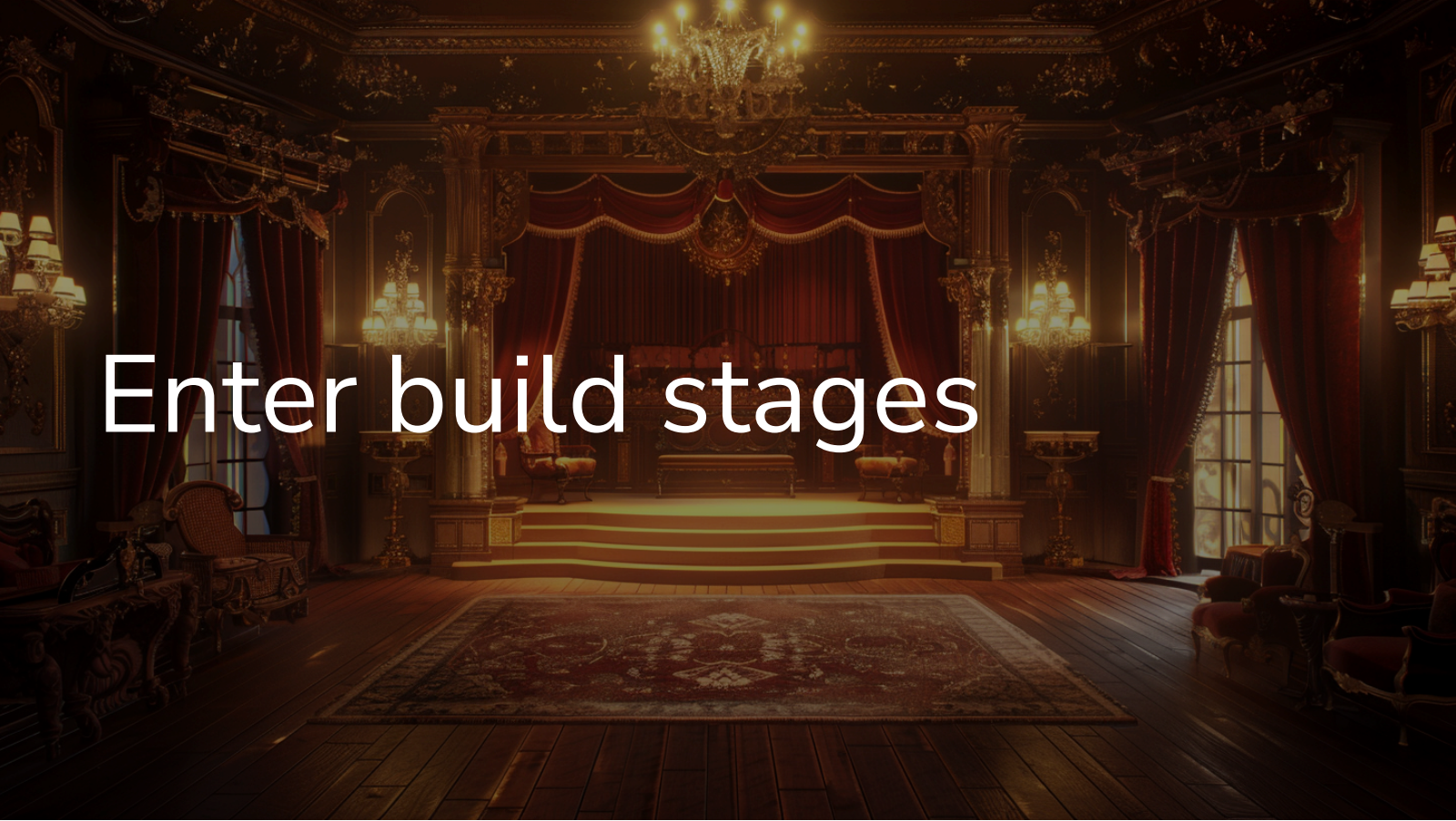
Saved 500+ MB

But the build is not clean now :(

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But the build is not clean now :(

We have to build everything outside, what if environment affects the build?

A highly detailed, ornate theater stage set. The stage is framed by a grand, classical archway supported by columns. Red velvet curtains hang from the top of the arch. A large, multi-tiered chandelier hangs from the ceiling, casting a warm glow. The stage floor is made of dark wood, and a large, patterned rug is laid out in the foreground. The walls are decorated with intricate carvings and sconces. The overall atmosphere is one of grandeur and elegance.

Enter build stages

Multi-stage builds

Holy grail of pure builds

- Allow clean builds
- Allow optimal packaging
- Allow different base images

Staged example

```
1 FROM bellsoft/liberica-runtime-container:jdk-musl as builder
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3 COPY . /app
4 RUN cd /app && ./gradlew build -xtest
5
6 FROM bellsoft/liberica-runtime-container:jre-slim-musl as runner
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8 COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
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3	d6eb71a5df		58.80MiB	COPY file:8d96bee
4	c6ede8dac2	bellsoft/liberica-runtime-container:jdk-21.0.3_10-musl	98.32MiB	15 BUNDLE_TYPE=;
5	<missing>		0B	ENV JAVA_HOME="/l
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That's all folks!

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Or is it?

What's these 59 MiB?

We have to pull 59 MiB of petclinic on every small commit!

Is there a way to optimize it?

Layers!

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6 FROM bellsoft/liberica-runtime-container:jre-slim-musl as optimizer
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8 COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
9 WORKDIR /app
10 RUN java -Diarmode=tools -jar /app/app.jar extract --layers --launcher
```

- Build image

Layers!

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- Build image
- Introduce new "optimizer" stage

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- Build image
- Introduce new "optimizer" stage
- Extract the jar to layered structure
- Copy layers

Layers!

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6 FROM bellsoft/liberica-runtime-container:jre-slim-musl as optimizer
7
8 COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
9 WORKDIR /app
10 RUN java -Diarmode=tools -jar /app/app.jar extract --layers --launcher
```

- Build image
- Introduce new "optimizer" stage
- Extract the jar to layered structure
- Copy layers

Layers

Because Spring Boot jar is complex!

```
1  Usage:
2    java -Djarmode=tools -jar my-app.jar
3
4  Available commands:
5    extract      Extract the contents from the jar
6    list-layers  List layers from the jar that can be extracted
```

Layers

```
1  app
2  |— application
3  |   |— BOOT-INF
4  |       |— classes
5  |           |— application-mysql.properties
6  |           |— application-postgres.properties
7  |           |— application.properties
8  |           |— banner.txt
9  |           |— db
10 |           |— ...
11 |           |— org
```

Layers

```
1  app
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Layers

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10 |   |   |   |— ...
11 |   |   |   |— org
```

Layered image structure

1	ID	TAG	SIZE	COMMAND
2	618743f6a2	layers:latest	2.96MiB	COPY dir:e0faa63b9654
3	9cca3273f0		0B	COPY dir:acd0d0ac1f7c
4	fe123ee16e		382.56kiB	COPY dir:01225d3c4ef0
5	dede9bac3d		57.69MiB	COPY dir:755815d928fc
6	65e3fb4cbf		0B	ENTRYPOINT ["java" "c
7	7130fa5864	bellsoft/liberica-runtime-container:jre-slim-musl	126.80MiB	18 CDS=no DESCRIPTION
8	<missing>		0B	ENV JAVA_HOME="/usr/1
9	<missing>		0B	ENV LANG=en_US.UTF-8
10	<missing>		0B	LABEL org.opencontainers

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9	<missing>		0B	ENV LANG=en_US.UTF-8
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- Dependencies: ~57.7MiB

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- Dependencies: ~57.7MiB
- Launcher: 382.56kiB

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- Application: ~3MiB

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8	<missing>		0B	ENV JAVA_HOME="/usr/1
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10	<missing>		0B	LABEL org.opencontainers

- Dependencies: ~57.7MiB
- Launcher: 382.56kiB
- Application: ~3MiB

Together: ~61MiB

61.0MiB > 58.8Mib!



What are we optimizing?

Pull size!

Pull size here is usually only around 3MiB!

We just reinvented how the BellSoft's buildpack works!

And it is amazing

Diversion: buildpacks

<https://paketo.io/>

Trivial usage:

```
1  /usr/sbin/pack build petclinic \  
2    --builder bellsoft/buildpacks.builder \  
3    --path . \  
4    -e BP_JVM_VERSION=21
```

Diversion: buildpacks

<https://paketo.io/>

Trivial usage:

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4   -e BP_JVM_VERSION=21
```

Result

1	ID	TAG	SIZE	COMMAND
2	2774e3f214	petclinic:latest	0B	Buildpacks Process Types
3	<missing>		1.44kiB	Buildpacks Launcher Config
4	<missing>		2.44MiB	Buildpacks Application Launcher
5	<missing>		59.17MiB	Application Layer
6	<missing>		729.21kiB	Software Bill-of-Materials
7	<missing>		97.87MiB	Layer: 'jre', Created by buildpack: bellsoft/buildpack
8	<missing>		200.00B	Layer: 'java-security-properties', Created by buildpack
9	<missing>		4.27MiB	Layer: 'helper', Created by buildpack: bellsoft/buildpack
10	<missing>		2.97kiB	
11	<missing>		7.48MiB	

Result

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Buildpacks

Why do we need them?

- Buildpacks transform your application source code into container images
 - The Paketo open source project provides production-ready buildpacks for the most popular languages and frameworks
 - Use Paketo Buildpacks to easily build your apps and keep them updated
- Reminds of s2i images
 - Build better than default
 - Spring-aware

Is this all?

We've optimized pull/push size, right?

Optimization is multidimensional

When startup time is more important...

There are several solutions for the Spring application startup time

1. Class Data Sharing (CDS)
2. Coordinated Restore at Checkpoint
3. Native Image

Class Data Sharing (CDS)

- When: Java 5 (!)
- Why: reduce the startup and memory footprint of multiple JVM instances running on the same host
- How: stores data in an archive

How does it help us?

Class Data Sharing (CDS)

- When: Java 5 (!)
- Why: reduce the startup and memory footprint of multiple JVM instances running on the same host
- How: stores data in an archive

How does it help us?

It does not! 🤖

Class Data Sharing (CDS)

- When: Java 5 (!)
- Why: reduce the startup and memory footprint of multiple JVM instances running on the same host
- How: stores data in an archive

How does it help us?

It does not! 🤖 But

Application Class Data Sharing

- When: Java 10
- Why: add application classes to the archive

And then:

- When: Java 13, JEP 350
- Why: allow addition of classes into the archive upon app exit!

And this helps! How?

How to use AppCDS with Spring?

- `-XX:ArchiveClassesAtExit=application.jsa` to create an archive
- `-Dspring.context.exit=onRefresh` to start and immediately exit the application

NB:

1. Use the same JDK
2. Use the same arguments

And even better!

Spring AOT

Add `-Dspring.aot.enabled=true`

Even more classes!!!

Practice

```
1 FROM bellsoft/liberica-runtime-container:jdk-musl as builder
2
3 COPY . /app
4 RUN cd /app && ./gradlew build -xtest
5
6 FROM bellsoft/liberica-runtime-container:jre-slim-musl as optimizer
7
8 COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
9 WORKDIR /app
10 RUN java -Djarmode=tools -jar /app/app.jar extract --layers --launcher
11
12 FROM bellsoft/liberica-runtime-container:jre-slim-musl as runner
13
14 ENTRYPOINT ["java", "org.springframework.boot.loader.launch.JarLauncher"]
15 COPY --from=optimizer /app/app/dependencies/ ./
16 COPY --from=optimizer /app/app/spring-boot-loader/ ./
17 COPY --from=optimizer /app/app/snapshot-dependencies/ ./
18 COPY --from=optimizer /app/app/application/ ./
```

Practice

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9 WORKDIR /app
10 RUN java -Djarmode=tools -jar /app/app.jar extract --layers --launcher
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12 FROM bellsoft/liberica-runtime-container:jre-cds-slim-musl as runner
13
14 ENTRYPOINT ["java", "org.springframework.boot.loader.launch.JarLauncher"]
15 COPY --from=optimizer /app/app/dependencies/ ./
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Practice

```
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8 COPY --from=optimizer /app/app/application/ ./
9 RUN java -Dspring.aot.enabled=true \
10     -XX:ArchiveClassesAtExit=./application.jsa \
11     -Dspring.context.exit=onRefresh \
12     org.springframework.boot.loader.launch.JarLauncher
```

Practice

```
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```

Practice

```
1 #omitted
2 FROM bellsoft/liberica-runtime-container:jre-cds-slim-musl as runner
3
4 ENTRYPOINT ["java", \
5             "-Dspring.aot.enabled=true", \
6             "-XX:SharedArchiveFile=application.jsa", \
7             "org.springframework.boot.loader.launch.JarLauncher"]
8 COPY --from=optimizer /app/app/dependencies / ./
9 COPY --from=optimizer /app/app/spring-boot-loader/ ./
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```

What does it cost ?

```
1  -r--r--r--      0:0      81 MB  |— application.jsa
```

Which is not small at all!

What does it cost ?

```
1    -r--r--r--    0:0    81 MB    |— application.jsa
```

Which is not small at all!

But you're trading pull speed for startup speed

How much faster?

Up to 50-60%

Pushing further with CRaC

CRaC: Coordinated Restore at Checkpoint

The CRaC (Coordinated Restore at Checkpoint) Project researches coordination of Java programs with mechanisms to checkpoint (make an image of, snapshot) a Java instance while it is executing. Restoring from the image could be a solution to some of the problems with the start-up and warm-up times. The primary aim of the Project is to develop a new standard mechanism-agnostic API to notify Java programs about the checkpoint and restore events. Other research activities will include, but will not be limited to, integration with existing checkpoint/restore mechanisms and development of new ones, changes to JVM and JDK to make images smaller and ensure they are correct.

<https://openjdk.org/projects/crac/>

In a perfect world it should be:

```
1  FROM bellsoft/liberica-runtime-container:jdk-musl as builder
2
3  COPY . /app
4  RUN cd /app && ./gradlew build
5
6  FROM bellsoft/liberica-runtime-container:jre-crac-slim as optimizer
7
8  COPY --from=builder /app/build/libs/spring-petclinic-3.3.0.jar /app/app.jar
9  WORKDIR /app
10 RUN java -Dspring.context.checkpoint=onRefresh -XX:CRaCCheckpointTo=/checkpoint -jar /app/app.jar
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12 FROM bellsoft/liberica-runtime-container:jre-crac-slim as runner
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14 ENTRYPOINT java -XX:CRaCRestoreFrom=/checkpoint
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But in reality

```
1  #12 6.051      Suppressed: java.lang.RuntimeException: Native checkpoint failed.
2  #12 6.051      at java.base/jdk.crac.Core.translateJVMExceptions(Core.java:114) ~[r
3  #12 6.051      at java.base/jdk.crac.Core.checkpointRestore1(Core.java:192) ~[na:na
4  #12 6.051      at java.base/jdk.crac.Core.checkpointRestore(Core.java:299) ~[na:na
5  #12 6.051      at java.base/jdk.crac.Core.checkpointRestore(Core.java:278) ~[na:na
6  #12 6.051      at java.base/javax.crac.Core.checkpointRestore(Core.java:73) ~[na:na
7  #12 6.051      at java.base/jdk.internal.reflect.DirectMethodHandleAccessor.invoke(
8  #12 6.051      at java.base/java.lang.reflect.Method.invoke(Method.java:580) ~[na:r
9  #12 6.051      at org.crac.Core$Compat.checkpointRestore(Core.java:141) ~[crac-1.4.
10 #12 6.051      ... 17 common frames omitted
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CRaC is hard!

Let's try to fix it with arcane magic

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And this is not all!

Did you hear about `buildx` ?

```
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2   '--allow-insecure-entitlement security.insecure' \  
3   --name insecure-builder
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```
1 docker buildx use insecure-builder
```

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```
1 docker buildx build \  
2   --allow security.insecure \  
3   -f Dockerfile.crac \  
4   -t pet-crac --output type=docker .
```

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1 docker run --rm -it --privileged pet-crac
```

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1 Restarting Spring-managed lifecycle beans after JVM restore
2 HikariPool-1 - Thread starvation or clock leap detected (housekeeper delta=1d19h37m42s102ms798µs806ns)
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4 Restored PetClinicApplication in 0.186 seconds (process running for 0.19)
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 2. `CAP_CHECKPOINT_RESTORE`
4. Container will run and stop
5. Commit the container like

```
docker commit container-id new-tag
```

Pros and Cons

Pros:

1. Does not require arcane magic
2. Works more predictably
3. Does not depend on unstable features
4. Does not require privileged containers in

Cons:

1. Organization-specific
2. Requires more steps

And now we have all
flavours of ultimate
docker images!

Quick summary?

1. Use layers for faster deployment
2. Use CDS for faster startup without many compromises
3. Use CRaC for a *lightning-fast* startup (with compromises)

Thank you!

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END